In [10]:
```
files = {
    "newholdingahftthtand.txt": "holding_hand",   # <-- this spelling
    "wavinghupdated.txt": "waving",
    "updchestresd.txt": "rest_chest",
    "ligtcupdatm.txt": "light_movement",
    "moderatehighchestahhttt.txt": "moderate_high_movement"
}
```

In [12]:
```
import os
os.listdir()
```

Out[12]:
```
['.ipynb_checkpoints',
 '0holdingtemp.csv',
 'armholding and waving.csv',
 'armholding and waving.txt',
 'chest rest updated for validation',
 'chestrestposition.txt',
 'contact presurrevariations.txt',
 'controledbreathing.txt',
 'highm.csv',
 'highm.txt',
 'holdingtemp.txt',
 'light mov chest for valid.txt',
 'lightmovementchest.txt',
 'moderate higgh chest movement.txt',
 'moderate high mov chest for valid.txt',
 'new holding aht on hand.txt',
 'Untitled.ipynb',
 'Untitled1.ipynb',
 'updated-restchestdata.csv',
 'updated-restchestdata.txt',
 'updatelightm.csv',
 'updatelightm.txt',
 'waving hand updated.txt']
```

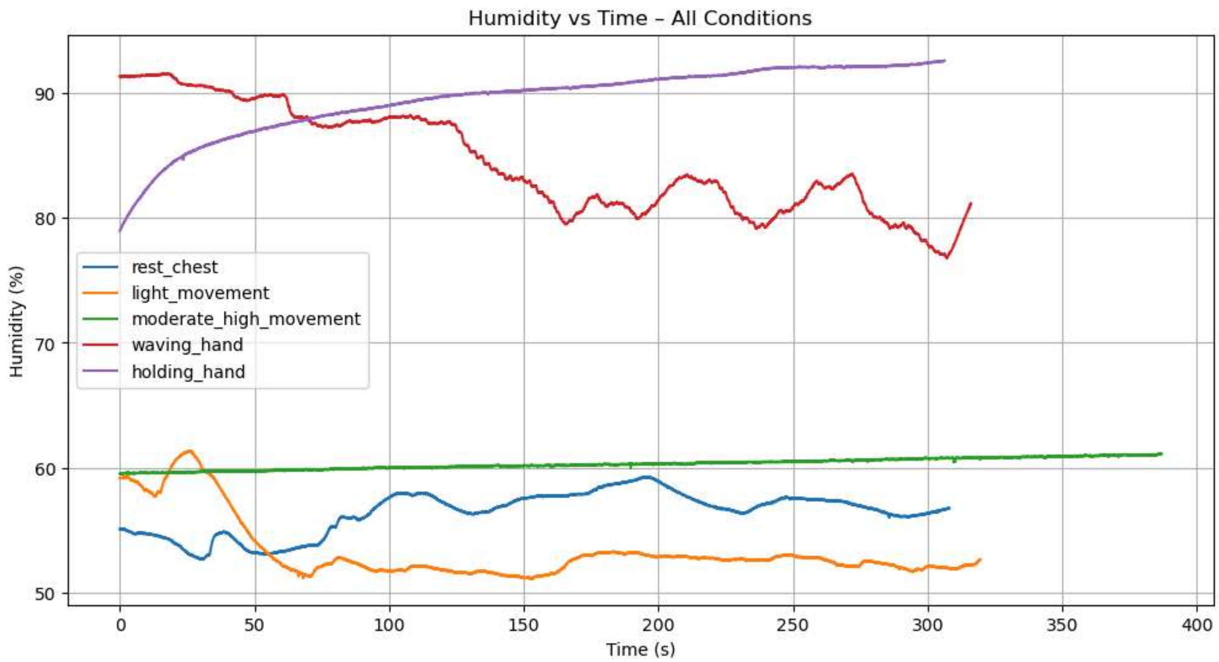
In [15]:

```python
import pandas as pd
import matplotlib.pyplot as plt

files = {
    "chest rest updated for validation.txt": "rest_chest",
    "light mov chest for valid.txt": "light_movement",
    "moderate high mov chest for valid.txt": "moderate_high_movement",
    "waving hand updated.txt": "waving_hand",
    "new holding aht on hand.txt": "holding_hand",
}

dfs = {}
summary_rows = []

for fname, label in files.items():
    df = pd.read_csv(fname, header=None, names=["t_ms", "tempC", "humidity"])

    # time in seconds starting at 0
    df["time_s"] = (df["t_ms"] - df["t_ms"].iloc[0]) / 1000.0
    df["condition"] = label
    dfs[label] = df

    # sampling stats
    dt = df["t_ms"].diff().dropna()
    mean_dt = dt.mean()
    fs = 1000.0 / mean_dt
    duration_s = df["time_s"].iloc[-1]

    summary_rows.append({
        "file": fname,
        "condition": label,
        "samples": len(df),
        "duration_s": duration_s,
        "mean_dt_ms": mean_dt,
        "fs_Hz": fs
    })

summary_df = pd.DataFrame(summary_rows)
summary_df
```

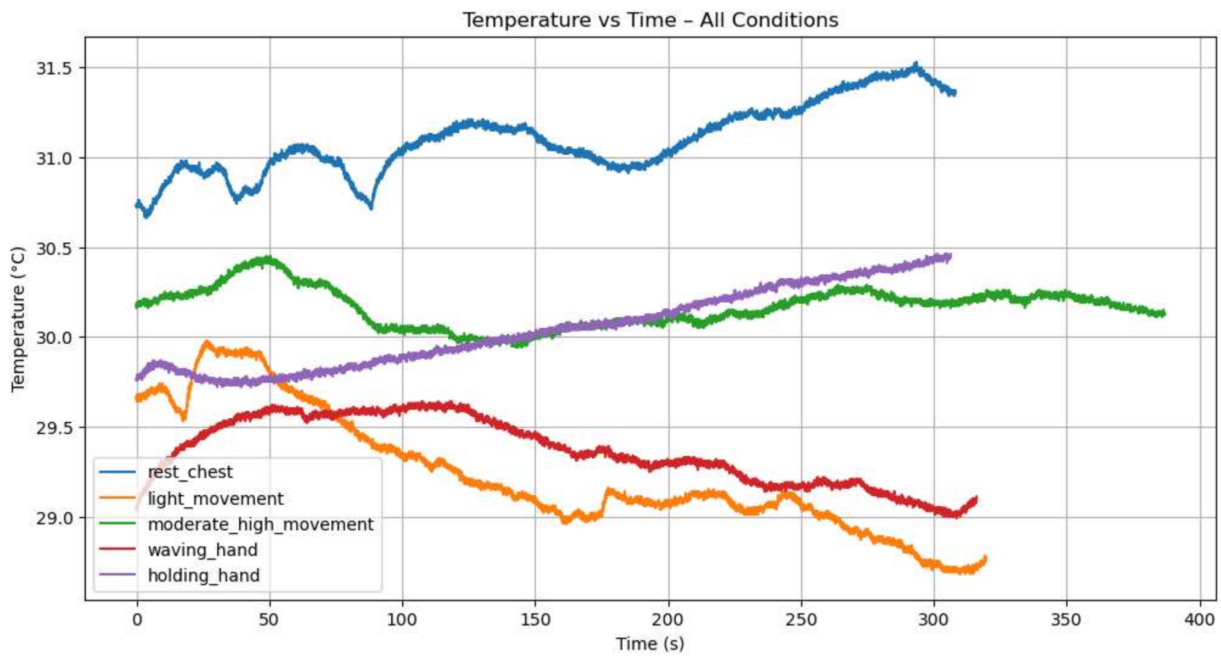
Out[15]:

	file	condition	samples	duration_s	mean_dt_ms	fs_Hz
0	chest rest updated for validation.txt	rest_chest	6161	308.00	50.0	20.0
1	light mov chest for valid.txt	light_movement	6391	319.50	50.0	20.0
2	moderate high mov chest for valid.txt	moderate_high_movement	7738	386.85	50.0	20.0
3	waving hand updated.txt	waving_hand	6322	316.05	50.0	20.0
4	new holding aht on hand.txt	holding_hand	6126	306.25	50.0	20.0

In [16]:
```
plt.figure(figsize=(12, 6))
for label, df in dfs.items():
    plt.plot(df["time_s"], df["humidity"], label=label)

plt.xlabel("Time (s)")
plt.ylabel("Humidity (%)")
plt.title("Humidity vs Time – All Conditions")
plt.grid(True)
plt.legend()
plt.show()
```



In [17]:
```
plt.figure(figsize=(12, 6))
for label, df in dfs.items():
    plt.plot(df["time_s"], df["tempC"], label=label)

plt.xlabel("Time (s)")
plt.ylabel("Temperature (°C)")
plt.title("Temperature vs Time – All Conditions")
plt.grid(True)
plt.legend()
plt.show()
```



In [ ]: